

UDIT KULKARNI

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Professional Summary

Software Engineer with a Master's in Computer Applications (Machine Learning), working across Laravel/PHP and Python/FastAPI. Experience building backend systems, RAG and voice applications, Text-to-SQL services, CRM/ERP platforms, high-concurrency APIs, and third-party integrations.

Core Skills

Backend: PHP, Laravel, Python, FastAPI, Magento 2, Drupal, Symfony, WebSockets

Frontend: JavaScript, ReactJS, HTML/CSS, Tailwind CSS

Data, Cloud & DevOps: SQL, MySQL, SQLite, Redis, AWS, Docker, Linux, Nginx, GitHub/GitLab CI, Jenkins

Professional Experience

Software Developer

Publicis Digital Experience

*Jan 2025 – Present
HINCOL & Hinduja Hospital*

HINCOL — Sales & Operations Platform

B2B Manufacturing

- Created a **multi-criteria Lead Scoring & Qualification engine** in Laravel using project type, volume, product specifications, margin and urgency, processing **800–1,500 enterprise leads/opportunities per month** across 5 national zones.
- Cut lead qualification from **~15 minutes to <1 second** and routing from **24–48 hours to real time**, saving an estimated **60–80 hours/month** of sales and leadership effort and reducing administrative vetting overhead by **~35%**.
- Designed a **Key Account Management (KAM) classification engine** for **~2,000–2,500 active accounts**, replacing quarterly spreadsheet reviews with event-driven and queued batch re-evaluation, plus a multi-stage approval state machine with AWS S3 audit attachments.
- Added a **Product Penetration & Cross-Sell matrix engine** to calculate customer whitespace deficits from sales orders, enabling sales teams to identify product-category gaps across customer accounts.
- Connected **Microsoft Graph API** to automate Outlook seminar calendar synchronization.
- Worked on a **77+ module Laravel platform** spanning leads, opportunities, RFQs, indents and credit governance, with SAP integrations for pricing, inventory, sales orders, bank guarantees and statements.

Hincol – Voice Assistant AI

- Architected and built **HincolBot**, a full-duplex voice application using FastAPI, WebSockets and Azure AI Foundry, reducing ERP/CRM data-entry time by **80%** across 5 national operating zones, from 10–15 minutes to **<2 minutes** per opportunity.
- Implemented bidirectional **24kHz PCM16 speech streaming** over authenticated WebSockets using GPT-Live-1 and browser Web Audio APIs, achieving **<200ms audio response latency** with real-time barge-in, interruption handling, Redis conversational memory and persistent session summaries.

Strategic Sales Agent

- Used a **LangGraph StateGraph** to manage multi-turn context, intent classification and dynamic slot filling, physically separating conversational reasoning from database execution and pausing for confirmation before mutations.
- Built **hybrid entity disambiguation** across **4,628 master vectors** spanning ~2,500 contractors/customers, ~1,500 locations, 20 product grades and 619 highway projects; fused Qdrant dense retrieval with lexical SQL search to achieve **98.4% accuracy across 500 conversational test cases**.
- Wrote a read-only **Text-to-SQL** service with vector schema discovery, read-only MySQL validation, sensitive-column masking and SSE streaming; added asynchronous Redis workflows and **LangSmith** observability for p50/p90/p99 latency, token usage, tracing and continuous AI evaluation.
- Added **HITL safeguards**: multilingual Zero-Delete protection, prompt-injection defenses, PII/schema validation and zone-scoped RBAC across 5 operating regions, requiring explicit confirmation before ACID database mutations.

Hinduja Hospital — Healthcare RAG & Conversational AI

Healthcare / HIS

- Built a **Healthcare RAG & voice assistant** using Python 3.11, FastAPI, Qdrant and LangChain for P.D. Hinduja Hospital (Mahim & Khar), integrating **260+ doctors, 90 specialties and 2 hospital units** for symptom triage, doctor discovery and real-time OPD booking.
- Combined dense retrieval and BM25 across **3,096 clinical chunks**, combining dense retrieval and BM25 to maximize recall across symptom language and exact medical entities, followed by **BAAI/bge-reranker-base** cross-encoder reranking for precision and clinical grounding.
- Tuned retrieval weights (70/30 dense-sparse) for symptom versus doctor/department queries, reducing a **~30-candidate pool to 6–12 grounded chunks** through reranking, threshold gating and token-budget allocation.
- Added an asynchronous **multi-turn state machine and HIS integration layer** supporting live slot discovery, SMS OTP authentication, BillDesk payment processing using JWS HMAC-SHA256 and automated hospital voucher generation; validated same-slot contention with **250 simultaneous booking attempts**.
- Ran a **320+ PyTest/PHPUnit regression suite**, achieving **96.6% pass rate for Mahim and 86.9% for Khar** in automated clinical and functional regression tests; end-to-end booking flows completed in **~45–60 seconds** in interactive testing.

Software Developer & Delivery Razorfish

Jan 2023 – Dec 2024
Disney+ Hotstar & Citroën

- Built the **Citroën Dealer Locator REST API**, aggregating real-time Stellantis/PSA global APIs, Citroën Advisor reviews and localized databases; supported a platform serving **1.5M+ monthly visitors** with **50+ RPS peak traffic** during national campaigns.
- Implemented geo-spatial dealer matching with real-time Haversine distance calculations, achieving **<350ms typical response latency**; deployed scalable containerized architecture using **Docker, AWS ECS/ECR, S3, CloudFront and Redis**.
- Delivered **12+ REST API modules** for the Citroën India digital platform using Drupal and PHP, including automated vehicle-data ingestion and synchronization across **1,000+ dealer locations**.
- Created a **Disney+ Hotstar campaign CMS** for drafting emailers and an in-house project-management tool for delivery tracking, invoicing and pending-payment management.
- Connected OTP verification, WhatsApp messaging and OpenID/OAuth authentication with third-party platforms, and improved page-load performance by **25%** through server-side caching and media-asset optimization.

Early Career

Software Developer & Project Management Intern Publicis Media

Jan 2022 – Jul 2022
View Certificate

- Built custom **Magento 2 modules** for an e-commerce POC with **PWA Studio**, using PHP, JavaScript and React; developed GraphQL APIs and worked with Elasticsearch and MySQL.
- Coordinated end-to-end project delivery by creating **Gantt charts and WBS structures**, tracking milestones, and coordinating communication across cross-functional teams while ensuring Agile practices were followed.
- Supported performance optimization, quality assurance and technical documentation covering configurations, customizations and development practices.

Education

Master's in Computer Applications (Machine Learning) Sardar Patel Institute of Technology (SPIT), Mumbai

2021 – 2022
View Degree **8.65/10**

Bachelor of Science in Computer Science

DG Ruparel College of Arts, Science and Commerce, Mumbai

2017 – 2020
View Degree **7.91/10**

Certifications & Recognition

- **Wow@Work Collective Impact Award** — HINCOL platform delivery View Award
- **Master RAG: Retrieval-Augmented Generation Systems** — Udemy, Sept 2026 Verify
- **Software Architecture & Design of Modern Large Scale Systems** — Udemy Verify
- **AI-900: Microsoft Azure AI Fundamentals** — Microsoft / Udemy Verify
- **GitHub Copilot for Developers: Code Faster With AI Agents** — Udemy Verify
- **Microsoft Technology Associate (MTA): Security Fundamentals** — Microsoft View Certificate